En Pensent — Worldwide Outreach Targets

Last Updated: February 19, 2026 **Current Metrics:** 2,804,090 chess games (EP 74.18% vs SF18 71.52%, +2.67pp, z>600), Market 36.1% 7-day directional (+15.7pp), *false_breakout* 60.0%, **7 validated domains** **Ingestion Rate:** Lichess API flowing | Chess.com saturated (saturation skip active) **Entity:** Canadian Corporation (pre-incorporation) | Owner: Alec Arthur Shelton

> ⚠️ ***INCORPORATION STATUS NOTE:*** *Several programs below require incorporation. Programs marked ✅ NO-CORP accept pre-incorporated founders directly. YC S26 deadline is* ***~April 2026*** *— applications likely open NOW.*

WHAT WE HAVE TO OFFER

En Pensent has built and validated a **universal pattern recognition architecture** — a single system ("the grid") that learns temporal-spatial patterns in ANY sequential domain:

1. **Chess** — 2,804,090 games, EP **74.18%** vs SF18 71.52% (+2.67pp, z>600, p≈0); golden zone 75.29% vs 72.78%
2. **Market Prediction** — 36.1% 7-day directional (+15.7pp over momentum baseline); *false_breakout* **60.0%** (n=919); 36,569 resolved predictions
3. **Battery Degradation** — 56.5%, 89.0% critical-state detection; 140 cells / 114,692 cycles (MIT-Stanford MATR)
4. **Chemical Fault Detection** — F1 **93.3%** vs 72.7% baseline on Tennessee Eastman Process (+20.6pp)
5. **Energy Grid** — 66.6% matching persistence baseline; 10,805 hourly records, 5 US regions
6. **Music (MAESTRO)** — 34.4% on 1,276 concert piano performances, 5.6M notes (+1.1pp over random)
7. **Nuclear Safety (NPPAD)** — Binary F1 **100%** (+11pp vs Bi-LSTM); 18-class **72.1%** / Macro-F1 50.0% (+31.4pp over NCC); NRC outage 62.8% (+6.4pp)

Projected accuracy with continued self-learning:

Volume	Chess EP (projected)	Market EP (projected)	Key Unlock
2.8M (now)	74.18% (+2.67pp edge)	36.1% 7-day directional	Current — 11-component fusion, per-archetype calibration
5M games	74.5–75.5%	38–42%	Denser archetype×phase interaction coverage
10M games	75–76%	42–48%	Per-player profiling, per-opening calibration
50M+ games	76–78%	50–55%	Approaching information-theoretic mid-game limits

Key differentiators:

- Same architecture across all 4 domains — no domain-specific neural networks
- Self-learning calibration from labeled data (no manual tuning)
- Runs on commodity hardware today, designed for photonic acceleration tomorrow

- 5,000× energy efficiency potential on silicon photonics vs GPU inference

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TIER 0: CANADIAN NON-INCORPORATED — NO INCORPORATION REQUIRED, APPLY NOW

These programs explicitly accept pre-incorporated founders. Apply before incorporating.

0A. Y Combinator (YC) — S26 Batch

- **What:** $500K for 7% equity. The most credentialing accelerator on earth. Handles incorporation during the batch.
- **Why us:** 7-domain validated system with 2.8M live predictions. No other applicant at S26 will have multi-domain running proof at this scale.
- **Deadline:** S26 applications typically close **April 2026** — apply NOW
- **Accepts non-incorporated:** ✅ Explicitly yes — they help you incorporate (Delaware C-corp) as part of the process
- **Accepts Canadians:** ✅ Yes — many YC companies are Canadian-founded
- **URL:** https://www.ycombinator.com/apply
- **Key pitch:** Universal Grid Portal — same 64-cell grid predicts chess outcomes, nuclear reactor faults, battery degradation, and market direction. 2.8M chess predictions (z>600), nuclear binary F1 100%, self-discovered z>3 universal anomaly constant. Software already executes what a photonic chip does optically.
- **Action:** Apply THIS WEEK

0B. Creative Destruction Lab (CDL) — AI / Photonics Streams

- **What:** Canada's premier deep-tech program. Not equity — objective-based mentorship by top entrepreneurs + access to investors at each sprint
- **Why us:** Universal photonic grid = CDL's exact focus. AI + hardware pathway.
- **Accepts non-incorporated:** ✅ Yes
- **Accepts Canadians:** ✅ Canadian (U of Toronto, Waterloo, UBC, Calgary, Montréal streams)
- **Streams:** AI (Waterloo/Toronto), Quantum (Waterloo), Climate
- **Deadline:** Next cohort intake typically March–April 2026
- **URL:** https://www.creativedestructionlab.com/apply/
- **Action:** Apply to CDL-AI stream (Waterloo or Toronto)

0C. Antler — Global Pre-Company Program

- **What:** ~$200K for ~10% equity. Designed specifically for pre-incorporation, pre-team stage.
- **Accepts non-incorporated:** ✅ Explicitly yes — Antler's entire model is helping you build the company from scratch
- **Accepts Canadians:** ✅ Yes (Toronto, NYC, London programs)
- **URL:** https://www.antler.co/apply
- **Action:** Toronto cohort — check next intake

0D. Entrepreneur First (EF)

- **What:** $100K–$250K for equity. Program for individuals before forming a company.
- **Accepts non-incorporated:** ✅ Yes — designed for individuals
- **Accepts Canadians:** ✅ Yes (London, NYC, Singapore programs)
- **URL:** https://www.joinef.com

0E. NEXT AI — Canadian AI Accelerator

- **What:** Canadian non-dilutive + small equity. Montréal/Toronto.
- **Why us:** Canadian AI company with 7 validated domains.
- **Accepts non-incorporated:** ✅ Yes
- **URL:** https://www.nextcanada.com/next-ai/

TIER 1: HIGHEST PROBABILITY — APPLY NOW (requires basic incorporation)

A. European Union Programs

1. EU Horizon Europe — EIC Accelerator

- **What:** €2.5M grant + €15M equity for breakthrough deep-tech startups
- **Why us:** Universal AI architecture across 4 domains = exactly their "deep tech" category
- **Eligibility:** Open to companies from associated countries. Canada has a framework agreement with EU for Horizon.
- **How:** Apply through EIC portal. Needs a 3-min video + 10-page proposal.
- **URL:** https://eic.ec.europa.eu/eic-funding-opportunities/eic-accelerator_en
- **Deadline:** Cut-off dates ~March, June, October annually
- **Action:** Check Canada-EU association status. If associated, apply directly. If not, partner with EU entity.

2. EU Horizon Europe — EIC Pathfinder Open

- **What:** Up to €3M for visionary research (early-stage, high-risk)
- **Why us:** Photonic universal grid = paradigm shift in AI compute
- **Eligibility:** Consortia of 3+ entities from different EU/associated countries
- **How:** Need EU academic partners (see faculty targets below)
- **URL:** https://eic.ec.europa.eu/eic-funding-opportunities/eic-pathfinder_en
- **Action:** Find 2+ EU academic partners, apply as consortium

3. UK Innovate UK — Smart Grants

- **What:** £25K–£2M for game-changing R&D projects
- **Why us:** AI + photonics innovation, cross-domain application
- **Eligibility:** Open to international collaboration (UK lead partner required)
- **How:** Partner with UK university or company
- **URL:** https://apply-for-innovation-funding.service.gov.uk/
- **Deadline:** Rolling (quarterly competition rounds)
- **Action:** Contact UK photonics partners (see faculty targets)

4. UK EPSRC (Engineering and Physical Sciences Research Council)

- **What:** £100K–£2M for research partnerships
- **Why us:** Photonic computing, AI architecture, signal processing
- **How:** Through UK university PI
- **Action:** Partner with Imperial College, Bristol, or Southampton photonics groups

5. Swiss SNSF (Swiss National Science Foundation)

- **What:** CHF 500K–5M for fundamental research
- **Why us:** Information theory foundations of the grid architecture
- **How:** Through Swiss university PI (ETH Zurich, EPFL)

B. Asian Programs

6. Japan JST (Japan Science and Technology Agency) — CREST / PRESTO

- **What:** ¥50M–300M (~$350K–$2M USD) for strategic basic research
- **Why us:** Japan is heavily investing in photonic computing and AI. JST CREST specifically funds "Photonic and Quantum Information Processing"
- **How:** Partner with Japanese PI (University of Tokyo, Keio, NTT Labs)
- **URL:** https://www.jst.go.jp/EN/

7. Singapore A*STAR — AI Research Grants

- **What:** S$500K–$5M for AI/ML research partnerships
- **Why us:** Singapore positions itself as AI hub. Universal grid = novel architecture.
- **Eligibility:** International collaboration welcome
- **How:** Partner with NUS, NTU, or A*STAR researchers
- **URL:** https://www.a-star.edu.sg/

8. Korea KIST / KAIST — AI & Photonics Programs

- **What:** ₩500M–5B (~$350K–$3.5M) for AI and photonic tech
- **Why us:** Korea investing heavily in AI chips and photonic computing
- **How:** Partner with KAIST or Seoul National University faculty
- **Action:** Contact KAIST photonics/AI groups

9. Abu Dhabi MBZUAI (Mohamed bin Zayed University of AI)

- **What:** Fully funded AI research partnerships + grants
- **Why us:** MBZUAI is building the world's first AI university. Universal grid is novel AI architecture.
- **How:** Direct outreach to faculty
- **URL:** https://mbzuai.ac.ae/

C. Australian/NZ Programs

10. Australia ARC (Australian Research Council) — Linkage Projects

- **What:** A$50K–$500K for university-industry R&D partnerships
- **Why us:** Cross-domain pattern recognition applicable to mining, energy, defence
- **How:** Partner with Australian university (ANU, UNSW, Sydney)
- **URL:** https://www.arc.gov.au/

TIER 2: ACADEMIC RESEARCH PARTNERSHIPS

Photonics & Optical Computing Labs (direct architecture relevance)

#	Institution	Lab / PI	Country	Why
1	**MIT**	Soljačić, Englund, Notaros	USA	KAN paper, photonic processor, top target
2	**Stanford**	Shanhui Fan (Photonics)	USA	Silicon photonics leader
3	**Princeton**	Prucnal (Photonic neural nets)	USA	Literally building photonic neural networks
4	**University of Oxford**	Harish Bhaskaran (Photonic computing)	UK	Phase-change photonic memory, Nature papers

5	**University of Bristol**	Mark Thompson (Quantum photonics)	UK	Quantum photonic chip company (PsiQuantum founder)
6	**University of Southampton**	Optoelectronics Research Centre	UK	World's largest photonics research group
7	**Ghent University / imec**	Photonics group	Belgium	Europe's silicon photonics hub
8	**ETH Zurich**	Jürg Leuthold (Photonics)	Switzerland	High-speed photonic processing
9	**EPFL**	Camille Brès (Photonics)	Switzerland	Nonlinear photonics, signal processing
10	**University of Tokyo**	Takagi Lab (Photonic computing)	Japan	Japan's leading photonic computing group
11	**NTT Research**	Photonics & Informatics Lab	Japan	Corporate research, photonic computing
12	**Université Laval**	COPL (Centre d'optique, photonique et laser)	Canada	Canada's premier photonics lab
13	**University of Waterloo**	IQC (Quantum computing)	Canada	Quantum photonic processors
14	**UBC**	Stewart Bhatt Lab (Photonics)	Canada	Silicon photonics, Canadian
15	**University of Toronto**	Joyce Poon (Photonics)	Canada	Silicon photonics, now at Max Planck
16	**Technion**	Mordechai Segev (Photonics)	Israel	Photonic topological insulators
17	**ANU**	Photonics group	Australia	Strong photonics + defence alignment

AI/ML Research Labs (novel architecture interest)

#	Institution	Lab / PI	Country	Why
18	**DeepMind**	Research partnerships	UK	Chess (AlphaZero), universal AI
19	**Mila (Montreal AI Institute)**	Yoshua Bengio	Canada	Fundamental AI research, Canadian
20	**Vector Institute**	Geoffrey Hinton (emeritus)	Canada	Toronto AI hub, Canadian grants pipeline
21	**CIFAR**	Pan-Canadian AI Strategy	Canada	Funds AI research across Canada

22	**Max Planck Institute**	ML group	Germany	Fundamental research, strong funding
23	**INRIA**	AI groups	France	French national AI research
24	**Alan Turing Institute**	AI research	UK	UK's national AI institute
25	**KAUST**	AI Initiative	Saudi Arabia	Massive research funding, AI focus

Chess-Specific Research (unique angle)

#	Institution	Contact	Why
26	**FIDE** (World Chess Federation)	Research committee	Governing body — would be interested in novel prediction methods
27	**Lichess**	Open-source team	Already using their data — partnership potential
28	**Chess.com**	Research / data team	Largest chess platform, potential data partnership
29	**ChessBase**	Research dept	Chess software leader — prediction integration

Quantitative Finance (market prediction angle)

#	Institution	Contact	Why
30	**Two Sigma**	Research partnerships	Quant fund, AI-driven trading
31	**Citadel / Citadel Securities**	Research team	Top quant fund
32	**Renaissance Technologies**	(very private)	Medallion Fund — ultimate quant validation
33	**AQR Capital**	Research publications	Publish research openly
34	**Man Group / Man AHL**	Oxford-Man Institute	Oxford partnership, open research
35	**WorldQuant**	University program	Global quant education + research
36	**Jump Trading**	Research team	Quant + tech firm

Industrial Process / Battery (domain validation angle)

#	Institution	Contact	Why
37	**Tesla**	Battery research / Autopilot	Battery degradation + sensor fusion
38	**CATL**	Battery R&D	World's largest battery manufacturer

39	**Samsung SDI**	Battery research	Battery degradation prediction
40	**Siemens**	Industrial AI / MindSphere	Chemical process monitoring
41	**ABB**	Process automation	TEP-style fault detection
42	**Honeywell**	Process Solutions	Chemical plant monitoring
43	**BASF**	Digital innovation	Chemical process optimization

TIER 3: VENTURE CAPITAL & ACCELERATORS

AI Infrastructure / Deep Tech VCs

#	Fund	Focus	Location	Why
44	**Lux Capital**	Deep tech, photonics	NYC	Invested in photonic computing companies
45	**DCVC (Data Collective)**	Deep tech + data science	SF	Computational science focus
46	**Radical Ventures**	AI companies	Toronto	Canadian, AI-focused, Hinton connection
47	**OMERS Ventures**	Canadian tech	Toronto	Major Canadian VC
48	**BDC Capital**	Canadian startups	Montreal	Canadian development bank
49	**Inovia Capital**	Canadian deep tech	Montreal	Invested in AI companies
50	**Playground Global**	AI/ML infrastructure	SF	Andy Rubin's fund, hardware+software
51	**Eclipse Ventures**	Industrial tech	SF	Manufacturing + deep tech
52	**Bessemer Venture Partners**	Enterprise AI	Global	Long-term technology bets
53	**Khosla Ventures**	Breakthrough tech	SF	Vinod Khosla funds unconventional tech
54	**Samsung NEXT**	Strategic investments	Global	Battery + AI + hardware
55	**Intel Capital**	Photonics + AI	Global	Direct photonic computing interest
56	**Google Ventures**	AI infrastructure	Global	Strategic alignment with AI compute
57	**Mubadala**	UAE sovereign wealth	Abu Dhabi	Deep tech, AI, massive capital

58	**Temasek**	Singapore sovereign wealth	Singapore	AI investments
59	**SoftBank Vision Fund**	AI at scale	Global	Large checks for transformative AI

Accelerators

#	Program	Focus	Location	Why
60	**Creative Destruction Lab**	Deep tech commercialization	Toronto (Canadian!)	Annual program, strong photonics/AI mentors
61	**Y Combinator**	General (strong AI track)	SF	Credibility + network (apply S26 or W27)
62	**Techstars**	Various programs	Global	AI-specific tracks
63	**NEXT AI**	AI startups	Toronto	Canadian AI accelerator
64	**Antler**	Deep tech	Global	Multiple locations, early-stage
65	**Station F**	Startup campus	Paris	EU hub, access to EU grants

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TIER 4: STRATEGIC INDUSTRY PARTNERSHIPS

Chess Platforms (data + distribution)

- **Lichess** — Open source, might host EP predictions as a feature
- **Chess.com** — Premium analytics integration potential
- **ChessBase** — Software integration for preparation tools

Financial Data Providers (market prediction)

- **Bloomberg** — Alternative data partnerships
- **Refinitiv/LSEG** — Market data provider
- **Quandl/Nasdaq** — Alternative data marketplace

Defence & Intelligence (Canadian entity advantage for Five Eyes)

- **DRDC (Defence Research and Development Canada)** — Direct government lab
- **CSE (Communications Security Establishment)** — Signals intelligence
- **Canadian Space Agency** — Satellite sensor fusion

IMMEDIATE ACTION PLAN

This Week

1. **IRAP** — Call NRC 1-877-994-4727, get ITA assigned (highest probability Canadian grant)
2. **Creative Destruction Lab** — Check application deadlines (Canadian, deep tech focus)
3. **Vector Institute / CIFAR** — Contact about research partnership + Canadian AI funding
4. **Mila** — Contact about fundamental AI research collaboration

Next 2 Weeks

5. **EU Horizon** — Check Canada association status, prepare EIC Accelerator application
6. **UK Innovate UK** — Find UK academic partner (Oxford, Bristol, or Southampton)
7. **SR&ED** — Contact SR&ED advisor for tax credit claim
8. **Radical Ventures / OMERS** — Prepare investor deck with current metrics

Next Month

9. **DARPA PICASSO** — Complete SAM.gov registration, submit application
10. **NSERC Alliance** — Contact Canadian university PIs (Laval, Waterloo, UBC)
11. **Mitacs** — Apply for R&D internship funding with university partner
12. **Japanese JST** — Contact University of Tokyo photonics group

Ongoing

- Monitor EU Horizon cut-off dates
- Watch for IDEaS defence calls
- Track ARPA-E OPEN BAAs
- Build relationships with target VCs

EMAIL TEMPLATES NEEDED

1. **Academic cold email** — for photonics/AI professors
2. **VC intro email** — for deep tech investors
3. **Chess platform partnership** — for Lichess/Chess.com
4. **Industrial partnership** — for Siemens/ABB/Honeywell
5. **Quant fund research inquiry** — for Two Sigma/AQR/Man Group
6. **EU grant partner inquiry** — for EU academic partners
7. **Canadian accelerator application** — for CDL/NEXT AI

This document identifies 65+ targets across 15 countries. Priority: Canadian programs first (highest probability), then EU Horizon (largest amounts), then strategic partnerships (revenue potential).